

Sunny Bagal

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Profile

Backend Software Engineer — Node.js, TypeScript, PostgreSQL, Redis. Built, benchmarked, and deployed two production systems end to end; every benchmark on this page is reproducible from a correctness-gated harness.

Skills

Languages: Python, JavaScript, TypeScript

Backend: Node.js, Express, WebSockets, Server-Sent Events (SSE), BullMQ, REST API design, JWT/OAuth 2.0, Prisma, Drizzle

Databases: PostgreSQL (SQL, pgvector), Redis (caching), MongoDB (NoSQL)

Infra & Tools: Docker, Docker Compose, Linux, Git, CI/CD (GitHub Actions)

Frontend: React, Next.js, Tailwind CSS

Professional Experience

IntelCorp Software India Pvt. Ltd., Software Engineering Intern (Technical Head) Jan 2026 – Apr 2026 | Navi Mumbai

- Led a 4-engineer team building GeoGrid, a geospatial lead-generation platform; owned the Node.js/Express backend and integrated teammates' modules (frontend, data enrichment, billing) behind a unified API.
- Devised the platform's core algorithm: recursive geospatial grid subdivision with ray-casting point-in-polygon filtering — any cell hitting Google Places' 60-result cap splits into sub-cells, recovering POIs far beyond the per-query limit.
- Designed a fixed-frame Redis grid-cell cache eliminating repeat Places calls — in production, region searches that previously took 10–15 minutes of API round-trips returned in 1–3 seconds; validated the caching pattern in a clean-room harness (100% call avoidance on identical repeats, 51.7% on 50% overlap, matching geometric overlap).

Projects

Linea — Real-time collaborative whiteboard, TypeScript, Next.js, Turborepo, PostgreSQL, Prisma, Canvas 2D Jul 2026

- Deployed at linea.sunnybagal.com; engineered the WebSocket sync layer from scratch (no Socket.IO): JWT-authenticated connections, room management, reconnection with exponential backoff, and a typed, Zod-validated message protocol.
- Designed state as a pure fold over an append-only, server-sequenced operation log — one reducer drives live sync, reload hydration, and read-only time-travel; undo appends compensating ops, never rewriting history.
- Diagnosed an $O(n^2)$ hydration fold via log-log scaling analysis (measured slope $2.12 \rightarrow 1.17$) and rewrote it as an $O(n)$ Map-based reducer verified op-for-op equivalent: 10,000-op replay 185 ms $\rightarrow$ 0.53 ms, 100,000-op boards hydrating in under 8 ms.
- Load-tested the sync path with a correctness-gated harness: 400 ops/sec sustained across 8 clients at ~5–6 ms p95 with zero loss, duplication, or reordering; identified the 800+ ops/sec ceiling as queueing on the per-op sequencing transaction.

Recall — Link Organizer, TypeScript, Node.js, PostgreSQL/pgvector, Redis, BullMQ, Claude API, React May 2026

- Deployed at recall.sunnybagal.com; BullMQ workers asynchronously fetch, summarize (Claude API), and embed each saved link into pgvector, keeping ingestion work off the request path.
- Implemented hybrid retrieval — pgvector HNSW + weighted tsvector/GIN full-text search, fused via reciprocal rank fusion — on a custom stack (no LangChain); migrating lexical search from ILIKE to GIN-indexed FTS cut hybrid p50 28.4 ms $\rightarrow$ 7.9 ms end-to-end.
- Built a 5-arm retrieval benchmark (1k–50k rows, recall@10, per-arm EXPLAIN) and shipped a transaction-scoped planner override — vector search p50 74.9 ms $\rightarrow$ 3.7 ms (20 $\times$) at 10k rows.
- Shipped an SSE-streamed RAG chat grounded in the user's saved links, with citation rendering.

NeuralCompress — Dual-Mode Image Compression System, PyTorch, CompressAI, DTCWT Jan 2025 – Jun 2025

- Designed a near-lossless image compression pipeline in PyTorch — Q-shift DTCWT with a reversible 8-block additive-coupling CNN trained via uniform-noise quantization simulation — achieving ~100 dB PSNR with reconstruction error below $1e-5$.
- Built a matched-quality benchmarking harness comparing a learned codec (CompressAI cheng2020-attn) against JPEG/WebP — 2.2 $\times$ smaller at equal ~33 dB PSNR (0.45 vs 0.98 BPP) — and shipped a custom binary format with zlib entropy coding via a Flask demo (huggingface.co/spaces/Sunny1110/image-compression).

Education

Bachelor of Technology (B.Tech.) in Computer Engineering
Narsee Monjee Institute of Management Studies

CGPA: 7.38/10

2022 – 2026 | Navi Mumbai